

Kelly Integration Spec for PolyAgora CTO Team

Part I — Classical Kelly Foundation

1. Objective

Kelly is **not** a prediction engine.

Kelly is a:

- capital allocation law,
- leverage sizing framework,
- geometric growth optimizer.

It answers:

Given a probabilistic edge, how much capital should be allocated?

2. Core Kelly Principle

For repeated multiplicative bets:

$$W_{t+1} = W_t(1 + fR_t)$$

where:

- W_t = capital,
- f = fraction allocated,
- R_t = return random variable.

Kelly maximizes:

$$\max E[\log W]$$

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Meaning:

maximize asymptotic geometric compounding.

3. Binary Kelly Formula

For:

- win probability p ,
- loss probability $q = 1 - p$,

- net odds b ,

optimal fraction:

$$f^* = \frac{bp - q}{b}$$

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Interpretation:

Condition	Kelly Output
no edge	$f = 0$
positive edge	allocate capital
stronger edge	larger allocation
negative edge	no allocation

4. Why Kelly Works

Kelly avoids:

- overbetting,
- ruin,
- volatility drag,
- exponential capital destruction.

Kelly optimizes:

- long-run growth rate,
- not one-step expectation.

This is essential in multiplicative systems.

5. Classical Quant Usage

Modern quant funds use Kelly-like sizing for:

- position sizing,
- leverage control,
- portfolio scaling,
- volatility targeting,
- risk budgeting.

Usually implemented as:

$$f_t \propto \frac{\mu_t}{\sigma_t^2}$$

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where:

- μ_t = expected edge,
- σ_t = volatility/risk.

But this assumes:

- stationary dynamics,
- stable transition laws,
- meaningful probabilistic estimation.

PolyAgora rejects these assumptions.

Part II — PolyAgora Kelly Architecture

6. Core Design Principle

Kelly MUST remain downstream from governance.

NOT:

Signal → Kelly

BUT:

Governance → Recoverability → Kelly

This preserves PolygonEye's semantic moat.

7. PolyAgora Runtime Flow

Runtime Pipeline

Market State → Regime Extraction → Survival Matrix → Fixed Star Block → MRTP :

8. Governance Before Capital

PolyAgora first determines:

Layer	Role
Survival Matrix	regime admissibility
Fixed Star	invariant structure
Block Logic	semantic coherence
M RTP	recoverable trajectory
Buffers/Ridges	transition tension
Veto	ordering contradiction
Zones	operational regime

Only afterward:

Kelly determines capital intensity.

9. Kelly Deformation Layer

This is the key PolyAgora innovation.

Classical Kelly:

$$f^*$$

PolyAgora Kelly:

$$f_t = f^* \cdot \Phi_t$$

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where:

$$0 \leq \Phi_t \leq 1$$

is the semantic governance deformation field.

10. Governance Deformation Field

Φ_t Components

Suggested structure:

$$\Phi_t = \varphi_Z \cdot \varphi_R \cdot \varphi_B \cdot \varphi_V \cdot \varphi_M$$

where:

Component	Meaning
φ_Z	Zone stability
φ_R	Recoverability
φ_B	Block coherence
φ_V	Veto proximity
φ_M	MRTP trajectory quality

11. Example Operational Mapping

Runtime State	Kelly Behavior
Stable Local Star	0.8–1.0 Kelly
Strong admissible corridor	0.6–0.8 Kelly
Ridge transition	0.25–0.5 Kelly
Buffer instability	0.1–0.25 Kelly
Boundary rupture	near zero Kelly
Veto condition	Kelly = 0
New emerging block	exploratory micro-Kelly

12. Recoverability-Adjusted Kelly

This is likely the deepest innovation.

Traditional Kelly sizes by:

Edge Magnitude

PolyAgora sizes by:

Edge × Recoverability

Meaning:

even strong alpha may receive low capital if:

- maneuverability is collapsing,
 - admissible futures narrow,
 - semantic ordering destabilizes.
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13. MRTP Integration

MRTP acts as runtime steering governor.

It evaluates:

- future reachability,
- recovery potential,
- semantic corridor preservation.

Suggested MRTP modifier:

$$\varphi_M = \frac{\text{Recoverable Futures}}{\text{Total Futures}}$$

Conceptually:

- high future optionality → expand Kelly,
 - narrowing corridor → shrink Kelly.
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14. Kelly Near Boundaries

Critical design requirement:

Kelly MUST collapse near semantic boundaries.

Reason:

Near ridge/buffer rupture:

- transition laws destabilize,
- estimation error explodes,
- edge reliability collapses,
- path dependency dominates.

Thus:

$$\Phi_t \rightarrow 0$$

as:

- veto risk rises,
- admissibility narrows,
- block coherence weakens.

15. PolyAgora vs Classical Markov Funds

Classical Funds

Assume:

$$P(S_{t+1}|S_t)$$

fixed probabilistic transitions.

Kelly operates directly on statistical edge.

PolyAgora

Assumes:

- transition law itself may deform,
- admissibility changes dynamically,
- ordering consistency matters,
- recoverability matters,
- semantic topology governs edge validity.

Thus Kelly becomes:

topology-aware capital modulation.

16. Suggested CTO Implementation

Phase 1 — Simple Kelly Overlay

Implement:

$$f_t = \lambda \cdot \frac{\mu_t}{\sigma_t^2}$$

with:

- fractional Kelly,
- capped leverage,
- rolling estimates.

Then multiply by:

$$\Phi_t$$

from governance layer.

17. Suggested Initial Governance Inputs

Version 1:

Signal	Weight
Zone score	30%
Block stability	25%
MRTP recoverability	25%
Buffer tension	10%
Veto proximity	10%

18. Kelly Safety Constraints

MANDATORY:

Constraint	Purpose
max leverage caps	prevent estimation blowup
fractional Kelly	robustness
regime hysteresis	avoid oscillation
veto hard stop	zero exposure
drawdown governor	survival preservation
boundary suppression	avoid false convexity

19. Recommended Kelly Modes

Mode A — Conservative

$$f_t = 0.25f^* \Phi_t$$

Production-safe.

Mode B — Adaptive

$$f_t = \alpha_t f^* \Phi_t$$

where:

$$\alpha_t$$

depends on:

- realized regime stability,
 - recent steering success,
 - block persistence.
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Mode C — Exploratory

Used near emerging blocks:

micro-Kelly probing allocations.

Purpose:

- information acquisition,
- not alpha extraction.

This is highly aligned with MRTP logic.

20. Strategic Interpretation

Classical Kelly:

maximize growth under uncertainty.

PolyAgora Kelly:

maximize maneuverable growth under changing semantic geometry.

This is a fundamentally different architecture.

Kelly is no longer merely:

- leverage sizing.

It becomes:

semantic capital steering under recoverability constraints.